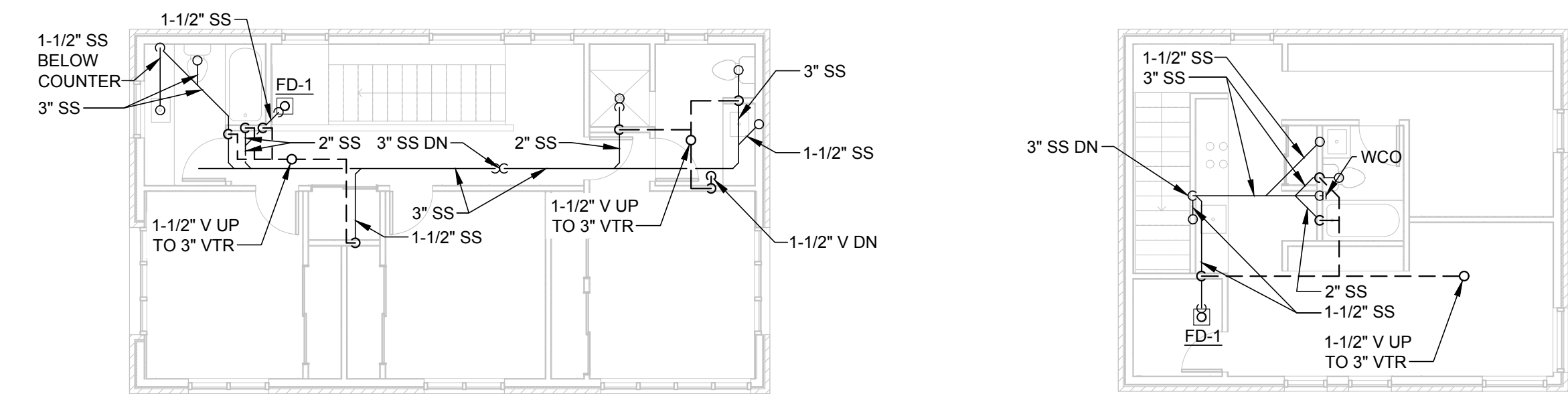


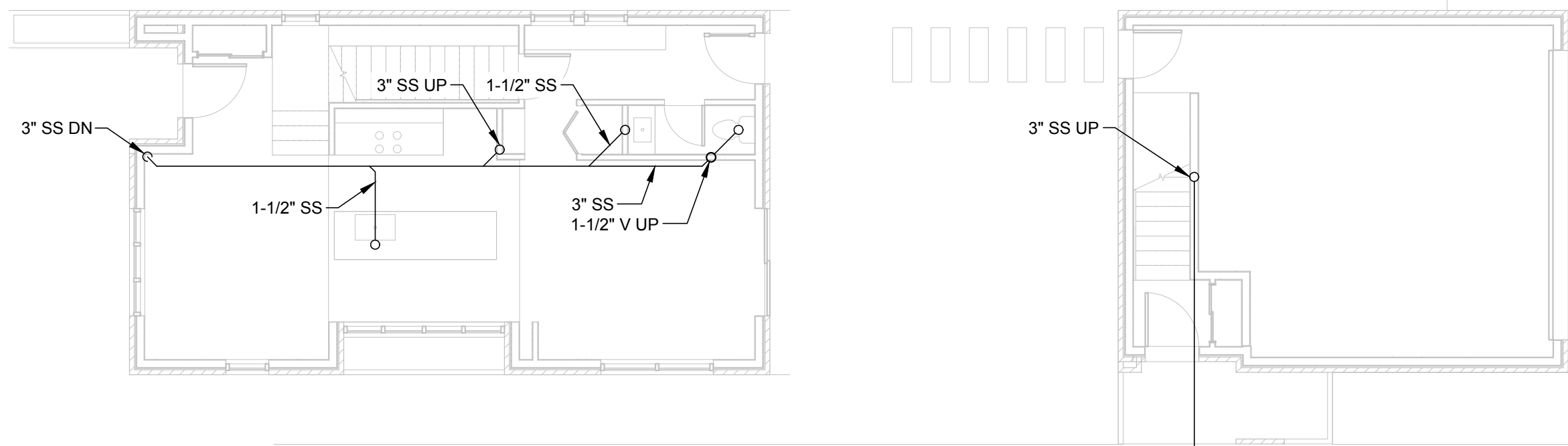
PLUMBING SPECIFICATIONS

- THE WORK SHALL BE EXECUTED IN STRICT CONFORMITY WITH BASE BUILDING SPECIFICATION AND WITH THE LATEST EDITION OF THE PREVAILING LOCAL PLUMBING AND BUILDING CODES AND ALL LOCAL REGULATIONS THAT MAY APPLY. IN CASE OF CONFLICT BETWEEN THE CONTRACT DOCUMENTS AND A GOVERNING CODE OR ORDINANCE THE MORE STRINGENT STANDARD SHALL APPLY.
- ALL PLUMBING WORK SHALL BE COORDINATED WITH ALL OTHER TRADES BEFORE PROCEEDING WITH INSTALLATION.
- PLUMBING CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS AND FOR PAYING RELATED FEES.
- ROUGH-IN DIMENSIONS OF TOILET FIXTURES MUST BE COORDINATED WITH GENERAL CONTRACTOR AND FIELD SUPERVISOR.
- INSTALL BALL VALVES ON ALL BRANCH SUPPLY LINES.
- PROVIDE ACCESS PANELS ON ALL INACCESSIBLE VALVES AND CLEANOUTS. ACCESS PANELS SHALL BE PROVIDED BY GENERAL CONTRACTOR. PLUMBING CONTRACTOR SHALL BE RESPONSIBLE FOR THEIR LOCATION. MOUNT SHUT-OFF VALVES NO HIGHER THAN 12'-8" AFF.
- ALL WORK SHALL BE PROPERLY TESTED, BALANCED AND CLEANED. PROVIDE A ONE YEAR WARRANTY FROM DATE OF FINAL INSPECTION ON ALL PARTS AND LABOR.
- FOLLOW PDI STANDARDS FOR WATER HAMMER ARRESTORS.
- ALL FIXTURES TO BE SUPPLIED & INSTALLED BY PLUMBING CONTRACTOR.
- PROVIDE BUILDING SERVICE LINE 5'-0" FROM BUILDING LINE, OR INSIDE BUILDING FROM OUTLET SIDE OF METER AS SHOWN ON CIVIL SITE PLAN. COORDINATE EXACT LOCATION WITH CIVIL DRAWINGS AND AT JOB SITE. PROVIDE BACKFLOW PREVENTER AS REQUIRED BY LOCAL AUTHORITIES.
- SANITARY SEWER PIPING SHOWN IS BASED ON 0.125"/FT FOR 3'-6" & 0.25"/FT FOR 2' 1/2" OR LESS FOR ALL PIPING. COORDINATE BUILDING SEWER LOCATION AND INVERT ELEVATION WITH CIVIL DRAWINGS.
- TRAP SEAL PRIMERS ARE TO BE PROVIDED ON FLOOR DRAINS/SINKS AS REQUIRED BY LOCAL BUILDING CODE OFFICIALS.
- VTRS SHALL BE LOCATED A MINIMUM OF 5' FROM ANY PARAPET WALL AND 10' FROM ANY RTU AIR INTAKE.
- FURNISH ALL MATERIAL, LABOR, EQUIPMENT, AND ACCESSORIES AS REQUIRED TO INSTALL COMPLETE PLUMBING AND HVAC PIPING SYSTEMS AS INDICATED ON MECHANICAL DRAWINGS AND IN THESE SPECIFICATIONS.
- SANITARY SEWERS
 - UNDERGROUND SANITARY SEWER PIPING SHALL BE SIZE 3" OR GREATER.
 - INSTALL SANITARY SEWERS, STACKS, VENTS, DRAINS, CLEANOUTS, ETC., AS INDICATED ON THE DRAWINGS.
 - SEWERS TO BE PITCHED A MINIMUM OF 1/8" PER FOOT.
 - CHANGES IN DIRECTION AND BRANCH CONNECTIONS SHALL BE MADE WITH APPROVED DRAINAGE FITTINGS COMPATIBLE WITH THE PIPING SYSTEM MATERIAL IN WHICH IT IS INSTALLED.
 - ALL FIXTURES AND SANITARY DRAINS SHALL BE VENTED AS INDICATED ON DRAWINGS AND IN ACCORDANCE WITH CODE.
 - VENTS ARE TO BE EXTENDED THROUGH ROOF AS INDICATED ON DRAINS AND FLASHED WITH 4 LB. LEAD WITH VENT FLASHING TOP TURNED DOWN TWO (2) INCHES MINIMUM INSIDE PIPE.
- BELOW GRADE SANITARY INSIDE BUILDING
 - SERVICE WEIGHT - CAST IRON PIPE ASTM A74-82 WITH ASTM C564-70 NEOPRENE COMPRESSION JOINTS OR NO-HUB WITH CLAMPS.
 - PVC-DWV PLASTIC ASTM D1785 WITH ASTM D2665 DWV SOLVENT WELD SOCKET FITTINGS.
- ABOVE GRADE SANITARY SEWER AND VENT
 - NO-HUB CAST IRON PIPE CISP1 1-301-78.
 - PVC-DWV PLASTIC ASTM D1785 WITH ASTM D2665 DWV SOLVENT WELD SOCKET FITTINGS (IN CONCEALED AREAS ONLY. PVC SHALL NOT BE INSTALLED IN CEILING PLENUMS).
 - SCHEDULE 40 GALVANIZED STEEL PIPE, ASTM A53-83 WITH CAST IRON SCREWED FITTINGS ANSI B16.22 1983.
- DOMESTIC WATER PIPING
 - INSTALL DOMESTIC WATER PIPING AS INDICATED ON DRAWINGS. INCLUDE ALL FITTINGS, VALVES, HANGERS AND OTHER ACCESSORIES. EXTEND DOMESTIC WATER PIPING TO ALL FIXTURES AND EQUIPMENT REQUIRED FOR COMPLETE INSTALLATION.
 - INCLUDE UNIONS, OR OTHER DISCONNECT MEANS, STOPS OR VALVES FOR ISOLATION OF FIXTURES AND EQUIPMENT. VALVES TO BE FULLY COMPATIBLE WITH PIPING FOR SERVICE INTENDED AS MANUFACTURED BY NIBCO, CRANE OR MILWAUKEE. INCLUDE HOSE OR DRAIN VALVES AT LOW POINTS WHERE FIXTURES CANNOT BE USED FOR DRAINAGE.
 - PROVIDE WATER HAMMER ARRESTERS OF TYPE AS INDICATED ON THE DOMESTIC WATER RISER DIAGRAM. THESE SHALL BE ASSE 1010 OR PDI-WH201, PISTON TYPE WITH PRESSURIZED METAL-TUBE CUSHIONING CHAMBER AS MANUFACTURED BY SIOUX CHIEF, AMTROL, JOSAM OR ZURN.
 - HANGERS ON INSULATED PIPE TO BE OUTSIDE OF INSULATION, SIZED ACCORDINGLY AND WITH A SUFFICIENT SADDLE TO PROTECT INSULATION.
- DOMESTIC WATER PIPING SHALL BE AS FOLLOWS:
 - ABOVE GRADE - TYPE "L" HARD COPPER ASTM B88-832 WITH WROUGHT COPPER FITTINGS ASTM B16.22 1980 AND NON-LEAD OR ANTIMONY SOLDER JOINTS.
 - BELOW GRADE - TYPE "K" SOFT COPPER WITHOUT SPLICES OR FITTINGS.
 - FLUSH, VENT AND SANITIZE ALL WATER PIPING WITH CHLORINE AS REQUIRED PER AWWA, LOCAL BUILDING DEPARTMENT AND HEALTH DEPARTMENT CODES.
 - DOMESTIC HOT AND COLD WATER PIPING UNDER CONCRETE FLOOR TO BE COVERED WITH SAND SO THAT PIPING WILL NOT BECOME EMBEDDED IN THE FLOOR SLAB.
 - ALL PIPING UNDER CONCRETE FLOOR SHALL BE TYPE "K" SOFT COPPER, CONTINUOUS. NO SPLICES OR FITTINGS WILL BE ALLOWED.
 - EXTREME CAUTION MUST BE TAKEN SO THAT NO COPPER PIPING AND INSULATION UNDER CONCRETE FLOORS BECOMES CRUSHED, CUT, SPLIT OR DEFORMED DURING THE POURING OF THE FLOOR SLAB.

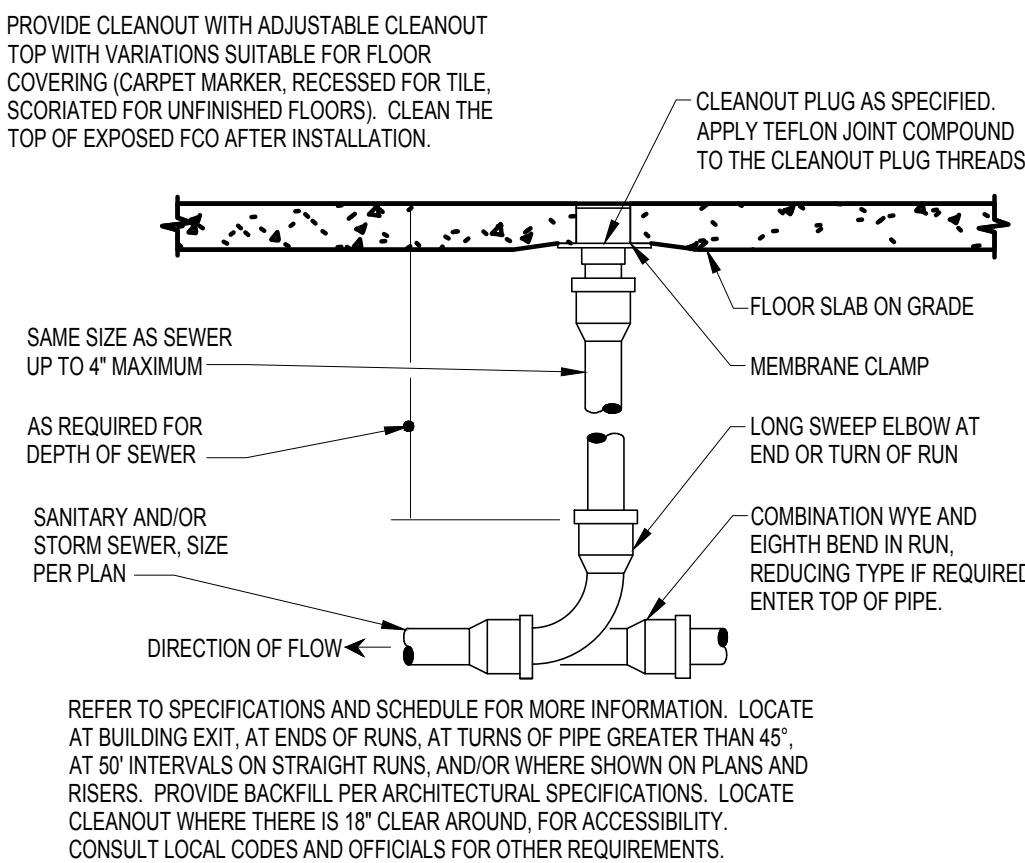
LEGEND	
	SANITARY SEWER
	DOMESTIC COLD WATER
	DOMESTIC HOT WATER
	CONDENSATE DRAIN
	PLUMBING VENT
	UNION
	PLUG COCK
	ELBOW - TURNED DOWN
	ELBOW - TURNED UP
	TEE - TURNED DOWN
	TEE - TURNED UP
	GATE VALVE
	BALL VALVE
	CHECK VALVE
	SHUT-OFF VALVE IN VERTICAL LINE
	T & P RELIEF VALVE
	FLOOR CLEANOUT
	YARD CLEANOUT
	WALL HYDRANT
	WALL / STACK CLEANOUT
FD	FLOOR DRAIN
DN	DOWN
VTR	VENT THRU ROOF
UG	UNDER GROUND
SS	SANITARY SEWER
V	PLUMBING VENT
HW	HOT WATER
CW	COLD WATER
CONT.	CONTINUATION
I.E.	INVERT ELEVATION
CD	CONDENSATE DRAIN



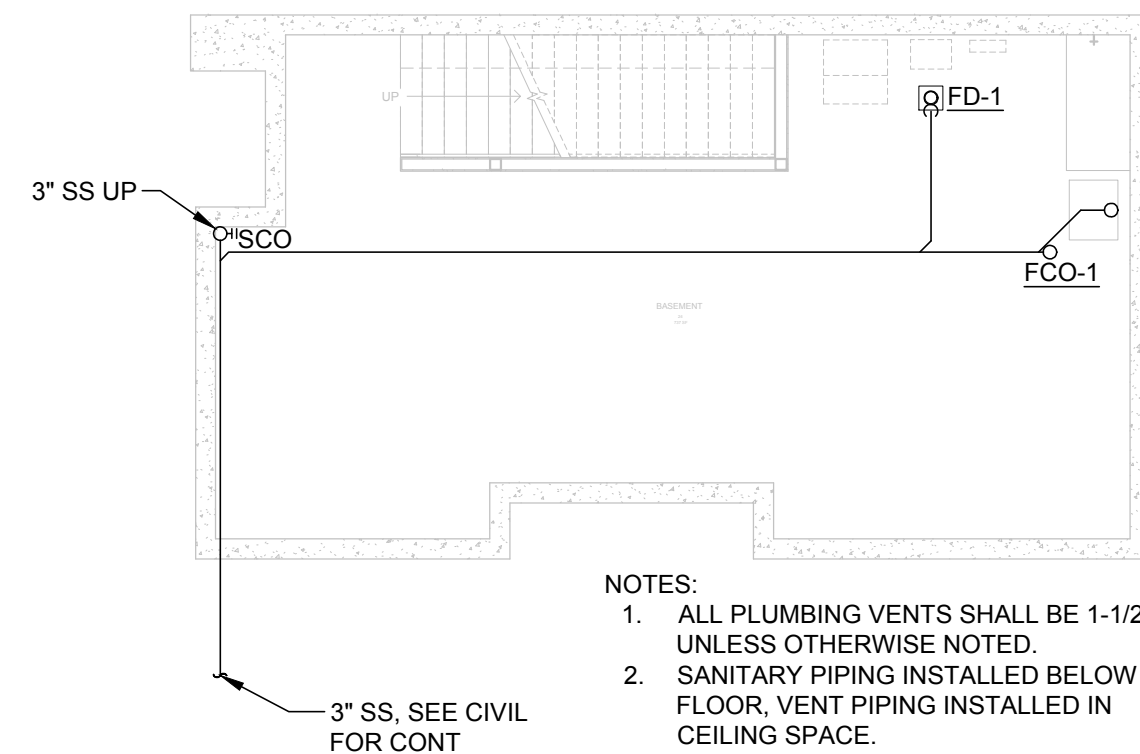
3 SECOND FLOOR WASTE/VENT PLAN
SCALE: 1/8" = 1'-0"



2 FIRST FLOOR WASTE/VENT PLAN
SCALE: 1/8" = 1'-0"



2 FLOOR CLEAN OUT DETAIL
Scale: No Scale



1 BASEMENT WASTE/VENT PLAN
SCALE: 1/8" = 1'-0"

GENERAL PLUMBING NOTES	
1	GENERAL PLUMBING NOTES APPLY TO ALL PLUMBING DRAWINGS.
2	THESE DRAWINGS ARE DIAGRAMMATIC & INDICATE THE GENERAL EXTENT OF THE WORK. PROVIDE PLUMBING SYSTEMS COMPLETE AND PER APPLICABLE CODES INCLUDING REQUIRED COMPONENTS, OFFSETS REQUIRED TO AVOID THE STRUCTURE, ETC.
3	REFER TO ARCHITECTURAL PLANS FOR SPECIFICATIONS OF AND EXACT LOCATIONS OF FIXTURES.
4	CONTRACTOR SHALL COORDINATE THE INSTALLATION OF PLUMBING AND PIPING WITH THE WORK OF ALL OTHER TRADES.
5	PIPING SHALL NOT BE LOCATED OVER ELECTRICAL EQUIPMENT/PANELS. PROVIDE REQUIRED CLEARANCE IN FRONT OF ELECTRICAL EQUIPMENT. PIPING SHALL NOT INTERFERE WITH ELECTRICAL EQUIPMENT CLEARANCE.
6	PIPING SHALL NOT BE INSTALLED IN A LOCATION THAT RESTRICTS THE ACCESS TO MECHANICAL DEVICES REQUIRING ACCESS.
7	CONTRACTOR SHALL PROVIDE SUPPLEMENTARY STEEL AS REQUIRED FOR THE PROPER SUPPORT OF ALL PLUMBING SYSTEMS.
8	HOT AND COLD WATER PIPING RUN-OUTS TO LAVATORIES AND SINKS SHALL BE 1/2" UNLESS NOTED OTHERWISE. REFER TO SPECIFICATIONS FOR FIXTURE CONNECTION SIZES.
9	PLUMBING VENT PIPING THRU THE ROOF SHALL BE LOCATED 10' FROM ANY FRESH AIR INTAKE LOCATION AND A MINIMUM OF 18" CLEAR FROM THE INSIDE FACE OF PARAPET.
10	PROVIDE CODE REQUIRED CLEARANCE FOR ALL CLEAN OUTS INSTALLED IN SANITARY WASTE AND VENT PIPING.
11	MINIMUM UNDERGROUND SANITARY SEWER PIPE SIZE SHALL BE 3".
12	SHUT-OFF VALVES SHALL BE LOCATED IN AN ACCESSIBLE AREA FOR EACH HW & CW BRANCH TO AN INDIVIDUAL FIXTURE.
13	ENGINEER OF RECORD SHALL RECEIVE SUBMITTALS FOR ALL PLUMBING FIXTURES. FIXTURES SHALL NOT BE INSTALLED WITHOUT APPROVAL.



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SEAL



THIS DRAWING SHALL NOT BE USED FOR CONSTRUCTION PURPOSES UNTIL THE SEAL AND SIGNATURE OF THE RESPONSIBLE REGISTRANT APPEARS ON THE DRAWING, AND PROPER PERMIT FORMS AND RELATED FEES ARE TRANSMITTED BY THE OWNER, OWNER'S AGENT OR CONTRACTOR TO THE AUTHORITY HAVING JURISDICTION.

CONSULTANT

CLIENT

Brian Mulloy

PROJECT

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Detroit, MI 48216

REVISIONS

06-26-2020 PERMIT

TITLE

PLUMBING WASTE &
VENT FLOOR PLANS

SHEET

P1